

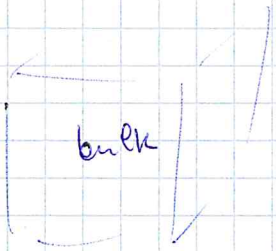
n_o

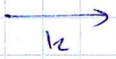


n_o

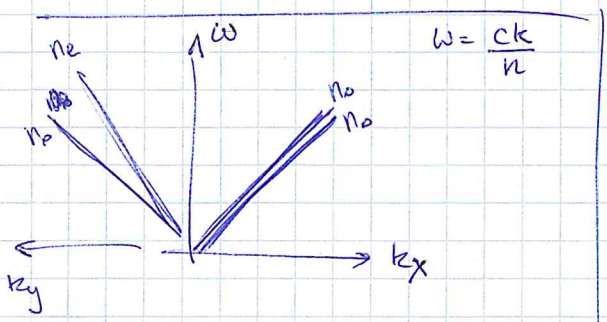


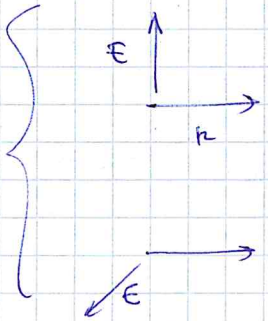
n_e





$k \parallel x$



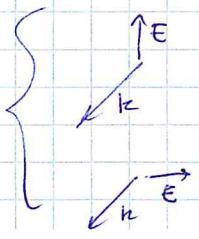


n_p

n_o

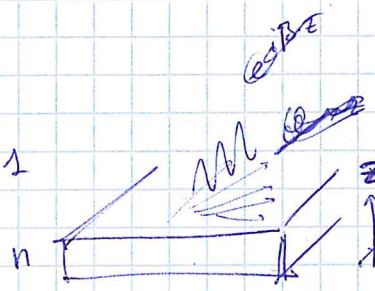


$k \parallel y$



n_o

n_e

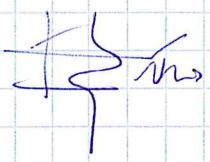


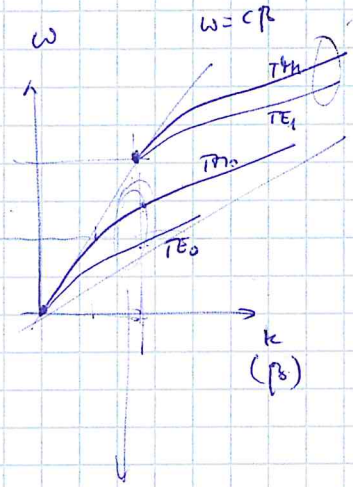
$e^{j\beta z}$

$\underline{E}(x,y,z) = e^{-j\beta \hat{s}\cdot\underline{r}} \cdot \Psi(z)$ $(x,y,0)$

$\beta = \frac{2\pi}{\lambda_0} \cdot n_{eff}$

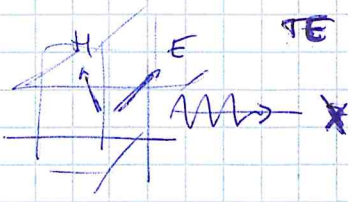
slab isotropo $(n_0\ n_0\ n_0)$

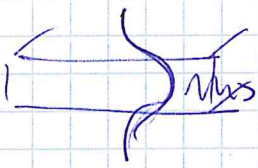


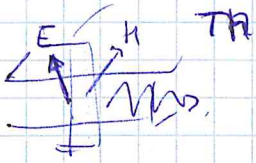


$\omega = \frac{c\beta}{n_0}$

form birifrigence



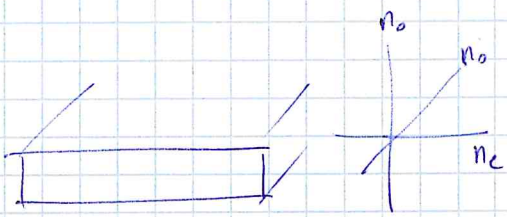


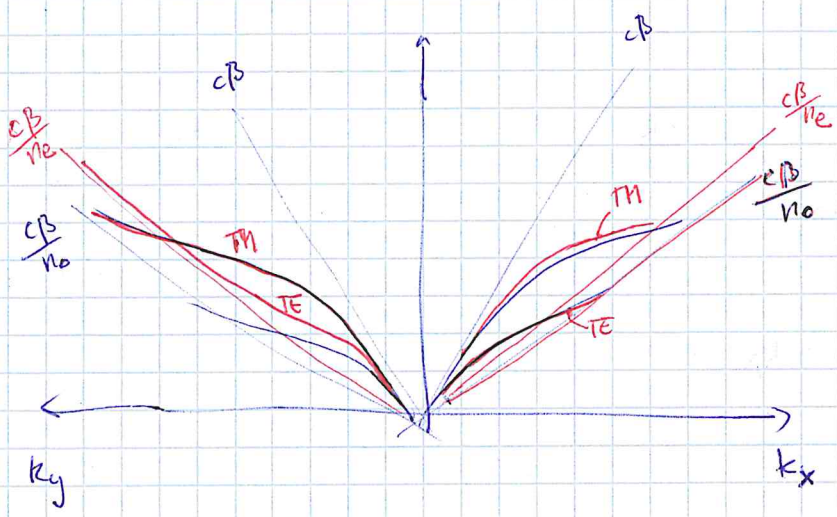


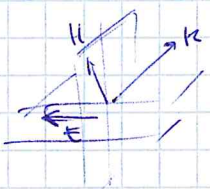


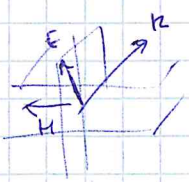
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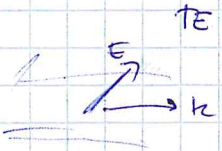
slab anisotropo

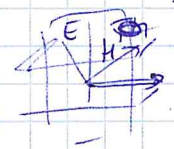


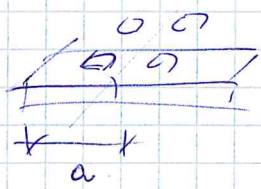


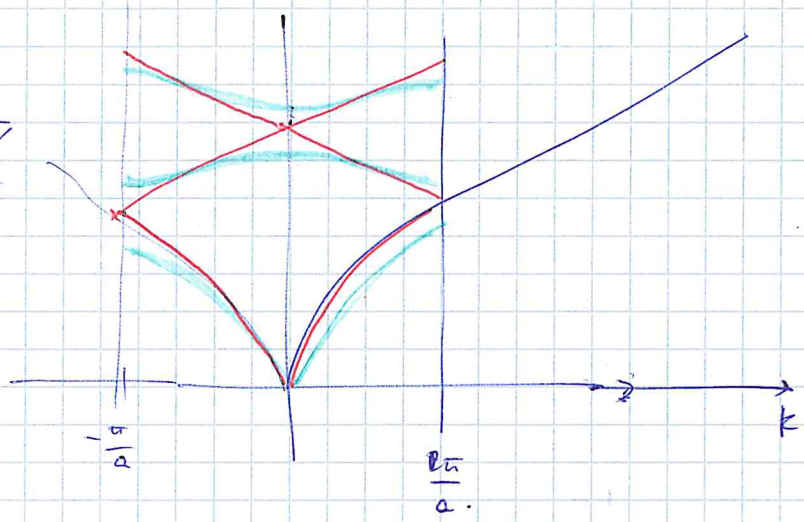


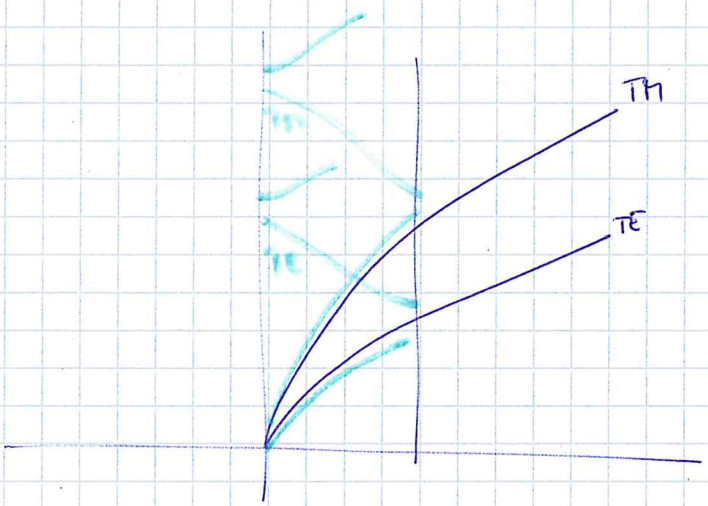


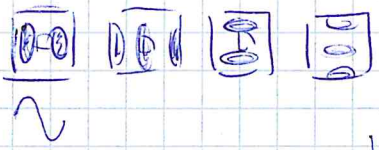


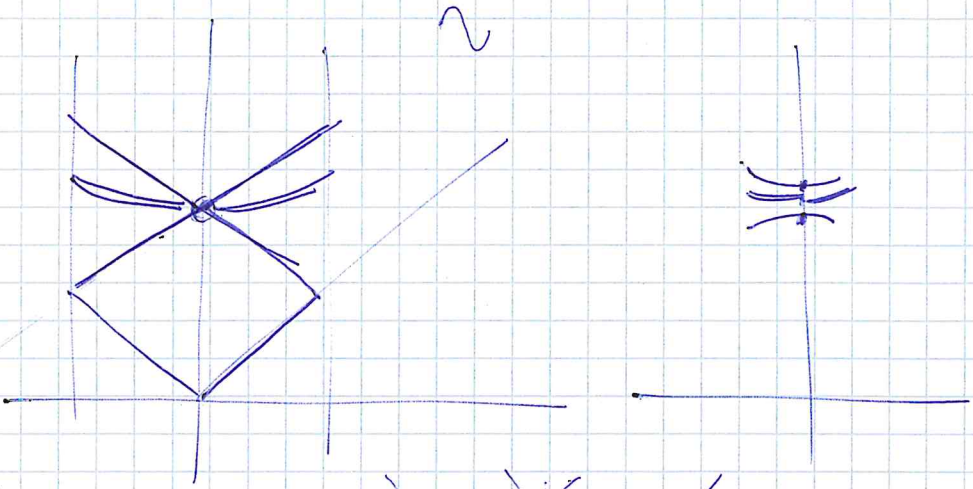


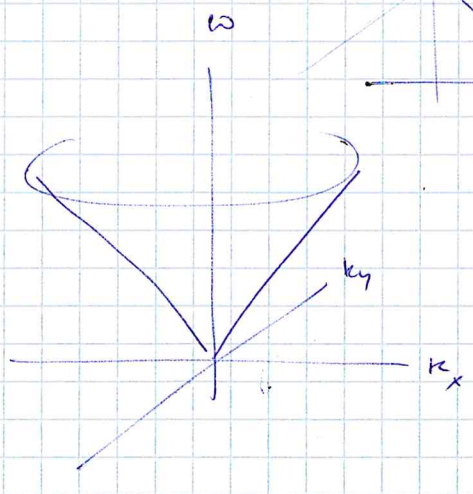

a

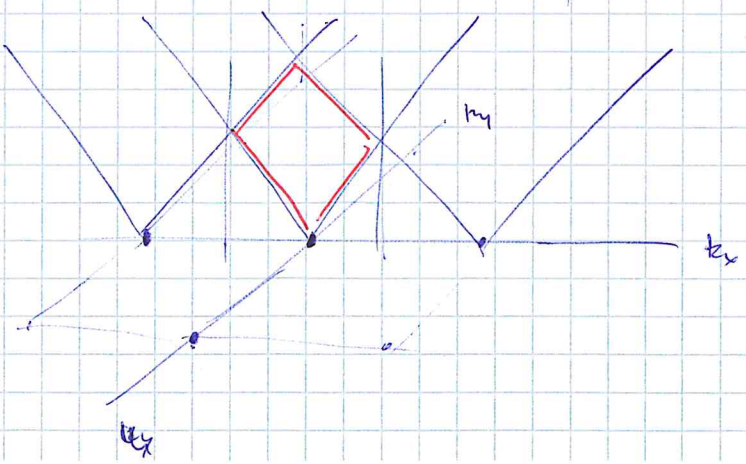

$-\frac{\pi}{a}$
$\frac{2\pi}{a}$
k

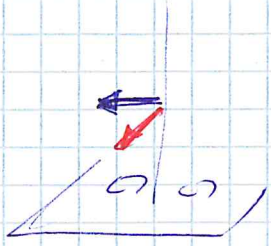

TM
TE

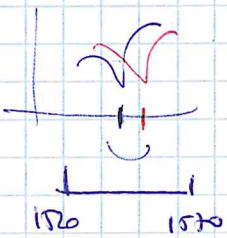


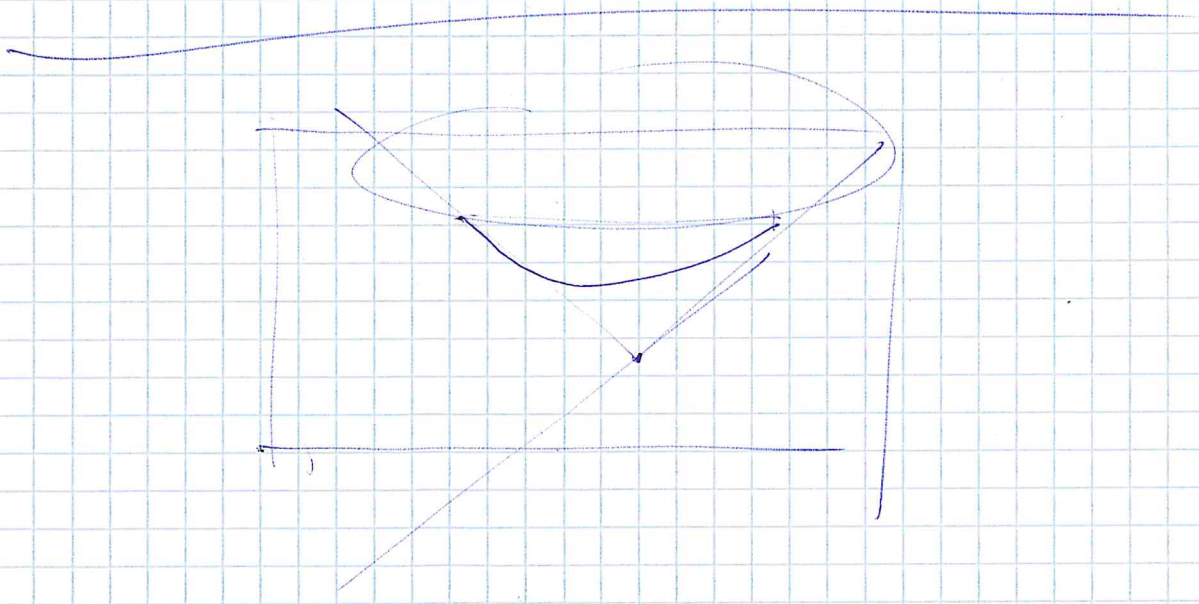



ω
k_y
k_x


k_y
k_x
k_y



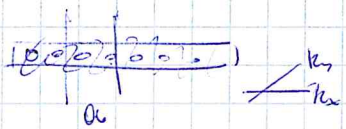

1520
1570

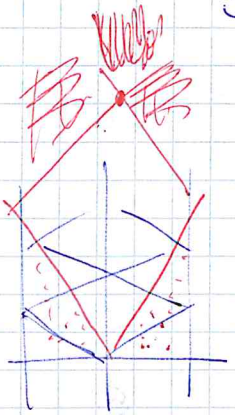


$e^{i k_\parallel \cdot r_\parallel} e^{i k_z z}$

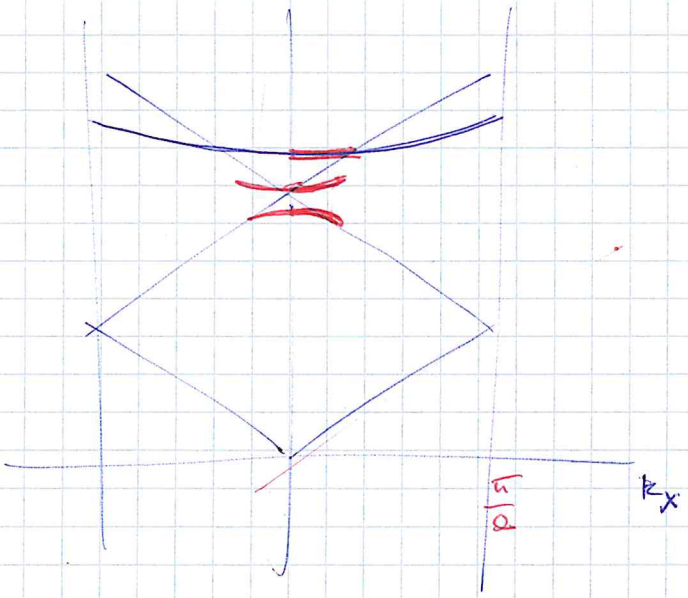
$|(k_x, k_y, k_z)|^2 = \frac{2\pi}{\lambda}$

$k_\parallel$

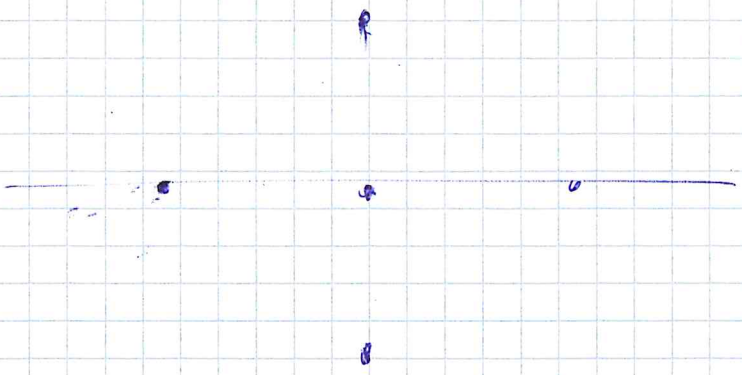




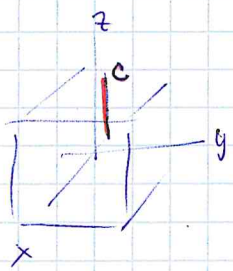
$\frac{2\pi}{\lambda_0} = \frac{2\pi}{a}$





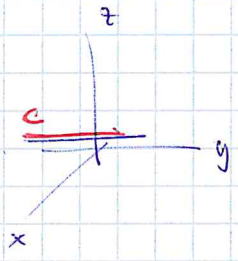


$$\varepsilon_{ij} = \varepsilon_{ij}\, T_{ii'}\, T_{dd'}$$



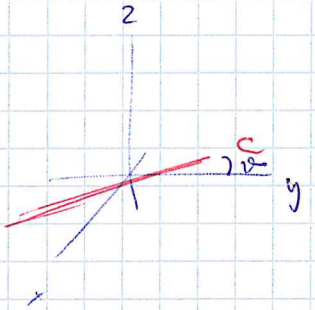
$$\varepsilon = \begin{pmatrix} \varepsilon_1 & 0 & 0 \\ 0 & \varepsilon_1 & 0 \\ 0 & 0 & \varepsilon_3 \end{pmatrix} \qquad \varepsilon_1 = n_o^2 \qquad \varepsilon_3 = n_e^2$$

$$T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

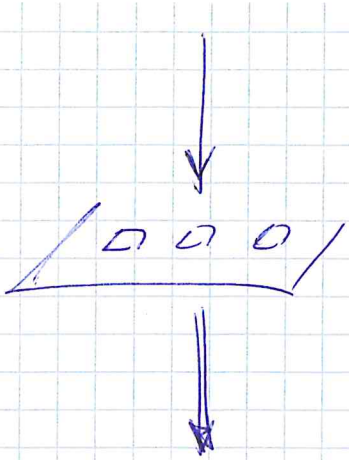


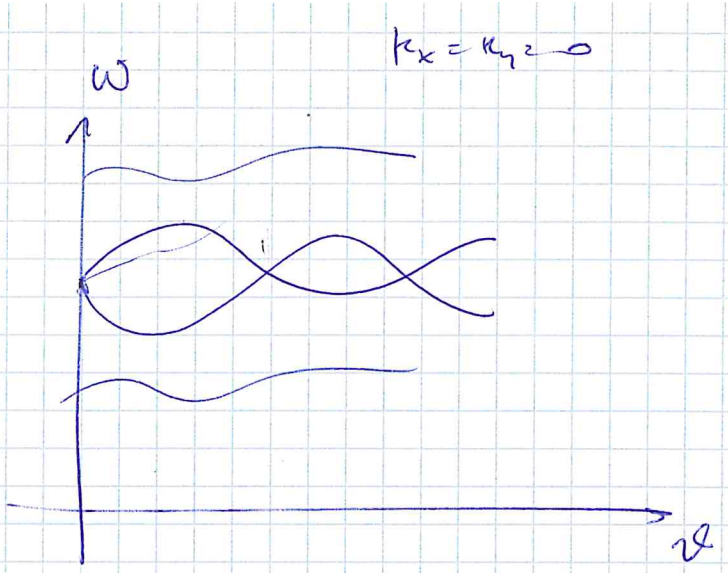
$$\varepsilon = \begin{pmatrix} \varepsilon_1 & 0 & 0 \\ 0 & \varepsilon_3 & 0 \\ 0 & 0 & \varepsilon_1 \end{pmatrix}$$

$$\underline{D} = \varepsilon\, \underline{E}$$

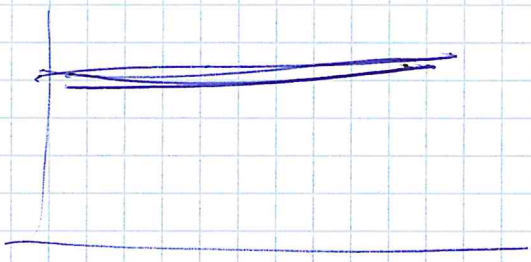


$$T = \begin{pmatrix} c_\theta & s_\theta & 0 \\ -s_\theta & c_\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$





$n_o \neq n_e$



$n_o = n_e$